

Researchers are studying a single-celled pathogen. Which experimental method would be LEAST effective for determining whether the pathogen is prokaryotic or eukaryotic?

- ☐ A. Analyzing the cells for the number of origins of replication [8%]
- ☐ B. Exposing the pathogen to a fluorescent antimitochondrial antibody [18%]
- ☐ C. Testing the pathogen for ribosome size [9%]
- ✔ ☒ D. Assessing the pathogen for the presence of pyruvate in the cytoplasm [63%]

Correct

62% answered correctly

Explanation:

	Prokaryotic cells	Eukaryotic cells
Cell types	Bacteria, Archaea	Animals, plants, fungi
Size	~1–5 μm	~10–100 μm
DNA	• Single circular chromosome • Double-stranded • Single origin of replication • Haploid • No telomeres • Located in cytoplasm	• Multiple linear chromosomes • Double-stranded • Multiple origins of replication • Diploid or greater • Telomeres • Located in nucleus
Cell structure & function	• No membrane-bound organelles • Cellular respiration in cytoplasm (aerobic & anaerobic) • Cell wall (eg, contains peptidoglycan in bacteria)	• Membrane-bound organelles • Cellular respiration in cytoplasm (anaerobic) & mitochondria (aerobic) • Cell wall (eg, contains cellulose in plants & chitin in fungi)
Ribosomes	70S ribosomes (composed of 50S & 30S subunits)	80S ribosomes (composed of 60S & 40S subunits)
Cell division	Binary fission	Mitosis (cell replication) & meiosis (gamete production)

Cells can be classified as either **prokaryotic** or **eukaryotic** based on specific cellular characteristics. Prokaryotes and eukaryotes differ in various traits, such as the number of origins of replication, presence of membrane-bound organelles, and genome location and morphology.

In the given scenario, scientists must identify a characteristic unique to either prokaryotes or eukaryotes that will allow them to classify the pathogen into one of the two groups.

However, **glycolysis** is a process that **occurs in** the cytoplasm of **both prokaryotic and eukaryotic** cells, and results in the synthesis of *pyruvate*. Therefore, the **presence of pyruvate** in the pathogen's cytoplasm **would not be a defining characteristic** that would help determine whether it is prokaryotic or eukaryotic.

(Choice A) In both prokaryotes and eukaryotes, DNA replication is initiated at the **origin of replication**. Prokaryotes have a single circular chromosome with a single origin of replication whereas eukaryotes have linear chromosomes with multiple origins of replication. By assessing the number of replication origins, researchers *can* determine whether a cell is prokaryotic or eukaryotic.

(Choice B) Mitochondria are the sites of aerobic respiration in eukaryotic cells but are absent in prokaryotic cells. An antimitochondrial antibody will bind only in the presence of eukaryotic cells, which would allow researchers to determine that the cell is eukaryotic and not prokaryotic.

(Choice C) Ribosomes translate mRNA into proteins in both prokaryotes and eukaryotes. However, prokaryotic and eukaryotic ribosomes are composed of **different subunits**. Eukaryotes have larger 80S ribosomes composed of a 60S and a 40S subunit whereas prokaryotes have smaller 70S ribosomes composed of a 50S and a 30S subunit. By assessing the size of the ribosomal subunits in the pathogen, researchers can identify whether it is prokaryotic or eukaryotic.

Educational objective:

Eukaryotic cells contain membrane-bound organelles and linear genomes with multiple origins of replication. In contrast, prokaryotic cells lack membrane-bound organelles and have circular genomes with a single origin of replication. However, both cells have ribosomes (80S in eukaryotes and 70S in prokaryotes) and can perform glycolysis.

Time spent:0 sec

Subject:
Biology

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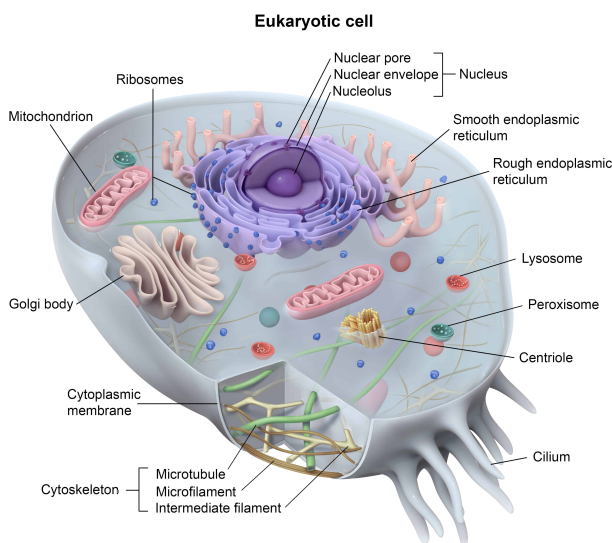
A student analyzing dividing cells of a certain type under the microscope finds that each daughter cell produced contains half the number of chromosomes as its parent cell. Given this, cells of this type would most likely contain all of the following EXCEPT:

- ☐ A. peroxisomes. [19%]
- ☐ B. nuclear pore proteins. [9%]
- ✓ ☒ C. single-stranded genomes. [62%]
- ☐ D. microtubules. [8%]

Correct

61% answered correctly

Explanation:



Eukaryotic cells can be distinguished from **other cells** by certain unique characteristics.

For example, a eukaryotic cell divides via **mitosis** or **meiosis**. Although mitosis produces daughter cells identical to the parent cell, meiosis yields four genetically unique daughter cells, each with *half* the genetic material of the parent cell. In contrast, a **prokaryotic cell** does *not* undergo mitosis or meiosis but instead divides to produce *identical* daughter cells via **binary fission**.

In this scenario, the student observed dividing cells that were producing daughter cells with half the number of chromosomes as the parent cell. Accordingly, this student must have been microscopically analyzing *eukaryotic cells* undergoing *meiosis*.

Eukaryotic cells are also unique in having **membrane-bound organelles** that make

up what is known as the *endomembrane system*. Because these organelles are filled with fluid, they are enclosed by a membrane to separate their interiors from the surrounding cytoplasm. The following are examples of these membrane-bound organelles:

- **Peroxisomes** are small organelles specialized to facilitate certain **oxidation-reduction reactions** involving peroxides as well as the metabolism of specific lipids. In addition, reactions in peroxisomes can facilitate detoxification of harmful substances within cells (**Choice A**).
- The **nucleus** is the storage site for DNA (ie, linear chromosomes) and where **DNA replication** and **transcription** (but *not* translation) of the **cell's genome** occur. Because **translation** takes place in the cytoplasm, the transcribed messenger RNA (mRNA) molecule must exit the nucleus. Consequently, nuclear membrane channels known as **nuclear pores** facilitate the passage of mRNA out of the nucleus and into the cytoplasm (**Choice B**). Nuclear pores, which also regulate the passage of certain proteins in and out of the nucleus, allow the compartmentalization of molecular processes.

Eukaryotic cells also have more solid (ie, not fluid-filled) structures that are not membrane bound (eg, ribosomes, cytoskeletal filaments). For instance, the filaments that make up the eukaryotic cytoskeleton include **microtubules** (**Choice D**) along with actin and intermediate filaments. The cytoskeleton supports the shape and internal structure of the cell and provides tracts for the transport of cellular cargo by **motor proteins**.

Single-stranded genomes, however, can be found in certain **viruses** but *not* in eukaryotic cells. Eukaryotic cells instead contain *double-stranded* DNA genomes.

Educational objective:

Eukaryotic cells can be distinguished by their unique characteristics. They have a genome with multiple linear chromosomes (double-stranded DNA). They also use meiosis as one mechanism of cell division (sexual reproduction), and they have membrane-bound organelles.

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